

Hospital ER Performance Analysis

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Tools: Excel, Power Query, Pivot Tables, Dashboard Design

Dataset: 10,100 Emergency Room visits

Executive Summary

The Business Question

A hospital with 10,100 annual ER visits needed to understand where patient experience was breaking down, why wait times were high, and how to improve operational efficiency. Four connected questions framed the investigation:

- What drives long patient wait times?
- What drives low patient satisfaction?
- Which departments face the highest operational pressure?
- What does the patient population look like, and when are they arriving?

What I Found

The analysis identified differences in wait times and patient satisfaction across departments, physicians, and shifts.

Finding 1: Wait times vary across departments and physicians

The average wait time was 31 minutes, with observed waits ranging from 1 to 199 minutes. Gastroenterology recorded the highest average departmental wait time, while physician averages also varied. Mark Smith recorded the highest average wait time at 36.28 minutes. Overall, 8.81% of patients waited more than 60 minutes.

These differences suggest that department and physician-level workflows are worth further investigation. The current analysis does not determine whether the variation is caused by staffing, patient case mix, scheduling, or another operational factor.

Finding 2: Satisfaction varies by department, but wait time alone does not explain the differences

Average satisfaction was 5.06 out of 10. Gastroenterology recorded the highest departmental average at 5.75, while Physiotherapy recorded the lowest at 4.56.

Patients waiting 61 - 120 minutes reported an average satisfaction score of 5.37, compared with 4.91 among patients waiting 0 - 15 minutes. In this dataset, longer waits were not consistently associated with lower satisfaction.

The available data does not explain the reason for this pattern. Other factors affecting patient experience would need further investigation.

Finding 3: Demand is seasonal and predictable.

The highest monthly visit volumes were recorded in May with 874 visits, December with 867, and March and August with 861 each. Monday recorded the highest day of week volume at 1,494 visits. Patients aged 31 - 45 were the largest age group, accounting for 2,778 visits. The gender distribution was nearly even, with 50.35% female and 49.65% male patients.

These patterns provide useful information for capacity planning, but additional years of data would be needed to confirm whether the monthly differences represent a recurring seasonal pattern.

What I Recommend

- Review departments and physicians with above-average wait times to identify differences in workflow, staffing, scheduling, and patient case mix.
- Investigate lower satisfaction scores in Physiotherapy and Emergency separately. Department-level patient characteristics, visit reasons, and operational processes should be reviewed before deciding on corrective action.
- Use monthly and day-of-week visit patterns to support staffing and capacity planning. May recorded the highest monthly volume, while Monday recorded the highest day-of-week volume in the dataset.
- Examine visit patterns and service needs among patients aged 31 - 45, the largest age group in the dataset, without assuming that one age group should define ER service design for all patients.
- Review physicians with lower average satisfaction scores alongside patient volume, department, wait time, triage level, and available patient characteristics before identifying coaching or performance interventions.

The Investigation

Dataset and Preparation

The raw dataset contained 10,130 Emergency Room visits with significant data quality issues that would have corrupted any analysis without correction. A full data quality audit was conducted before any analysis began.

Key issues found and resolved:

- 5,523 missing Department values, replaced with 'Unknown' to preserve row count
- 7,088 missing Satisfaction Scores, retained as blank rather than imputed, since absent feedback is not the same as poor satisfaction
- 16 negative wait times and 14 placeholder values of 999 minutes, 30 records removed (0.30% of dataset)
- Gender coded inconsistently as M / F / Male / Female / 'Maleale' / 'Femaleemale', standardized to Male / Female
- Triage levels represented as 1, L1, ESI 1, and Level 1, all meaning the same severity, standardized to Level 1 through Level 5
- Shift values coded as Morning / MORNING / Morning Shift, standardized to Morning / Afternoon / Night

- Date features engineered: Visit Year, Month, Day of Week, and Hour to enable time-series analysis
- Star schema built: 1 Fact table (Fact_ER_Visits) and 4 Dimension tables (Dim_Department, Dim_Physician, Dim_Diagnosis, Dim_ReferralSource)

Final cleaned dataset: 10,100 records, with the identified data quality issues documented and treated for analysis and dashboard development.

Dashboard Structure

Four linked dashboards were built in Excel to answer each business question:

Dashboard Focus	Key Metrics Key Visual	Business Question Answered
Executive Overview	Total visits, admission rate, readmission rate, avg satisfaction, avg wait	How is the ER performing overall?
Wait Time Analysis	Avg wait by dept, shift, triage, physician, and hour	Where are patients waiting longest?
Patient Satisfaction	Satisfaction by dept, shift, triage level, and wait group	What drives patient experience?
Patient Demographics	Age, gender, race, monthly volume, daily average	Who is coming to the ER and when?

Key Findings by Dashboard

Executive Overview

- 10,100 total ER visits in the dataset period
- 42.45% of ER visits in the dataset were recorded as admitted.
- 8.99% of visits were recorded as readmissions.
- Average satisfaction: 5.06 out of 10
- Average wait time: 31 minutes
- Night shift handles the highest patient volume; Monday is the busiest day of the week

Wait Time Analysis

- Average Wait: 31 minutes; longest: 199 minutes; shortest: 1 minute
- 8.81% of patients waited over 60 minutes, 889 people
- Gastroenterology has the highest departmental average wait time
- Afternoon shift produces the longest average waits; Night shift is shortest
- Average wait time was similar across triage levels in the dataset. This pattern is worth further investigation to understand how triage priority, patient condition, and ER workflow affected actual waiting times.
- Mark Smith has the highest average wait time among all physicians at 36.28 minute

Patient Satisfaction

- Gastroenterology: 5.75 highest department
- Physiotherapy: 4.56 lowest department
- Emergency: 4.71 second lowest, worth monitoring given volume
- 37.4% of patients rated their experience 7-10 (Highly Satisfied)
- Night shift scores highest satisfaction at 5.08; Morning shift lowest at 4.96
- Longer wait groups recorded higher average satisfaction scores in this dataset: 5.37 for 61 - 120 minutes and 5.36 for 120+ minutes, compared with 4.91 for 0 - 15 minutes. The available data does not explain the reason for this pattern.
- Bottom physician: Martinez, D. at 3.80 average 1.26 points below hospital average

Patient Demographics

- Average patient age: 45 years, consistent across all departments (43-48)
- Gender split: 50.35% Female (5,085), 49.65% Male (4,987)
- Largest age group: 31-45 at 2,778 visits
- Busiest months: May (874), March (861), August (861), December (867)
- Daily average: 28 ER visits per day
- Largest racial group: White (2,806), followed by African American (2,123) and Two or More Races (1,729)

Skills Demonstrated

Skill	Where Used	Evidence
Excel & Power Query	Data import and cleaning	Standardized 8 categorical fields; removed 30 invalid records; engineered 4 date features
Pivot Tables	All four dashboards	Built structured pivot tables across wait time, satisfaction, demographics, and physician dimensions
Data Cleaning	Pre-analysis audit	Documented and resolved missing values, invalid records, and inconsistent categories in a formal audit report
Dashboard Design	All four pages	Linked dashboards with KPI cards, bar charts, donut charts, and line charts across four analytical themes
Star Schema Modeling	Data preparation	Built 1 Fact table + 4 Dimension tables before dashboarding
Business Analysis	All sections	Framed four business questions; produced specific, prioritized recommendations per dashboard
Data Storytelling	This document	Organized the analysis around business questions, findings, and recommendations

About

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